Instructor: Dr. Natalia Chaudhry

Software Quality Engineering, SEF21

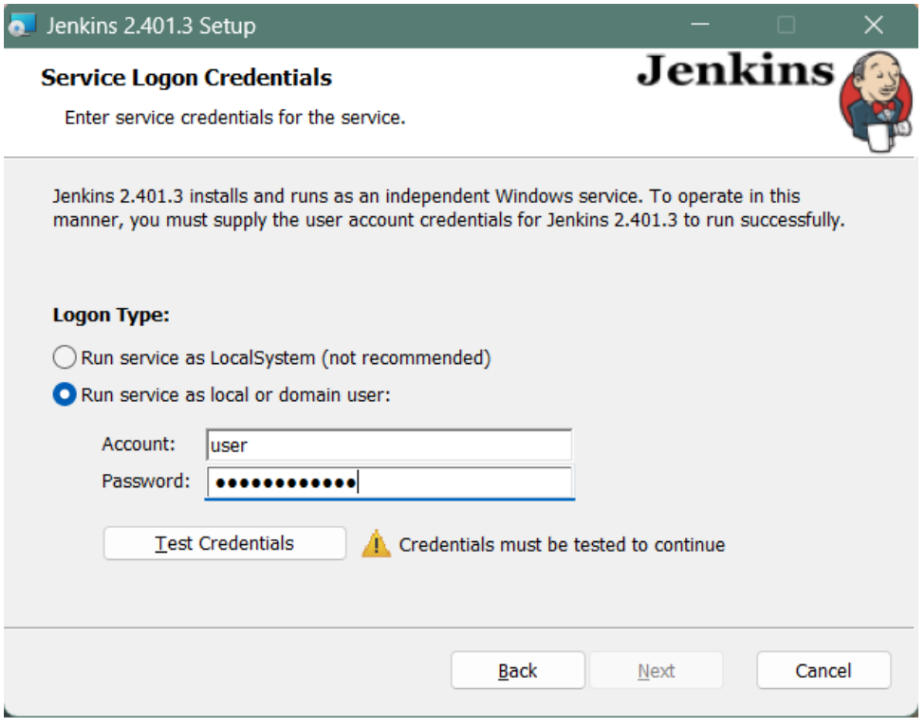
Dated: 23-10-2024

SQA Lab#2 : Exploring Jenkins for DevOps

Install Jenkins:

https://www.jenkins.io/download/#downloading-jenkins

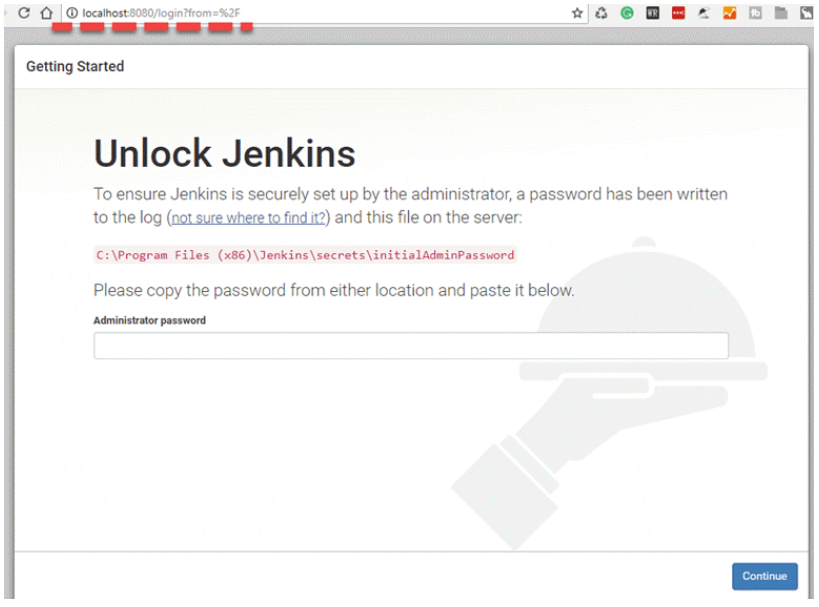
You can choose either of the logon type. Choose first option for simplicity.



After completing the Jenkins installation process, a browser tab will pop-up asking for the initial Administrator password.

To access Jenkins, you need to go to browse the following path in your web browser.

http://localhost:8080

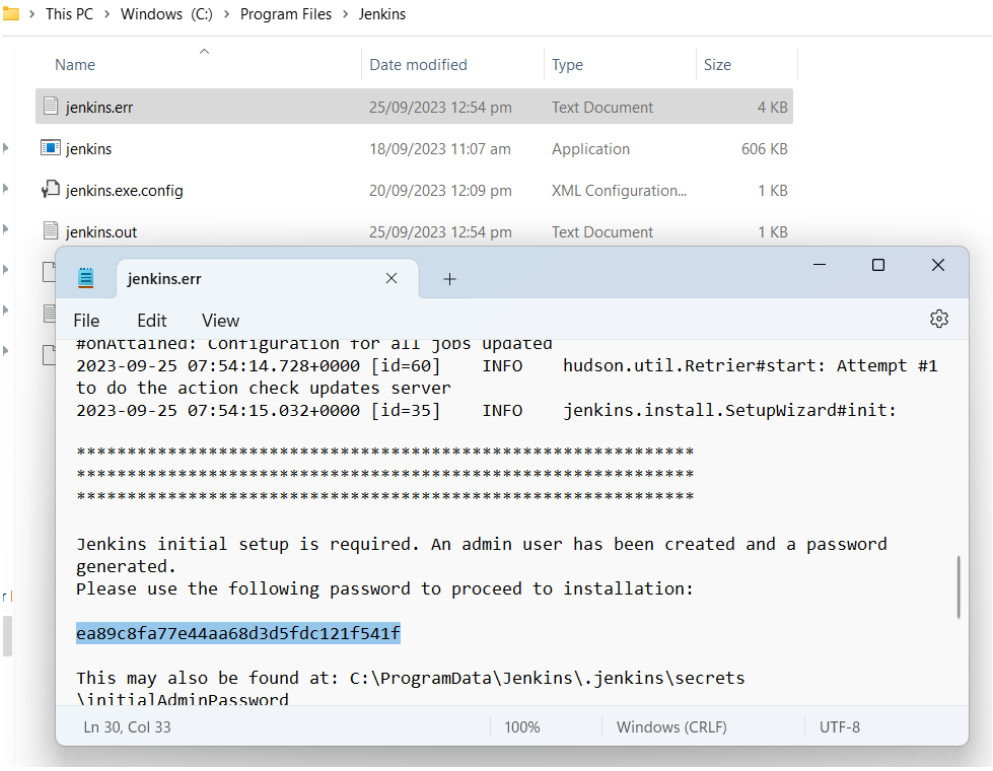


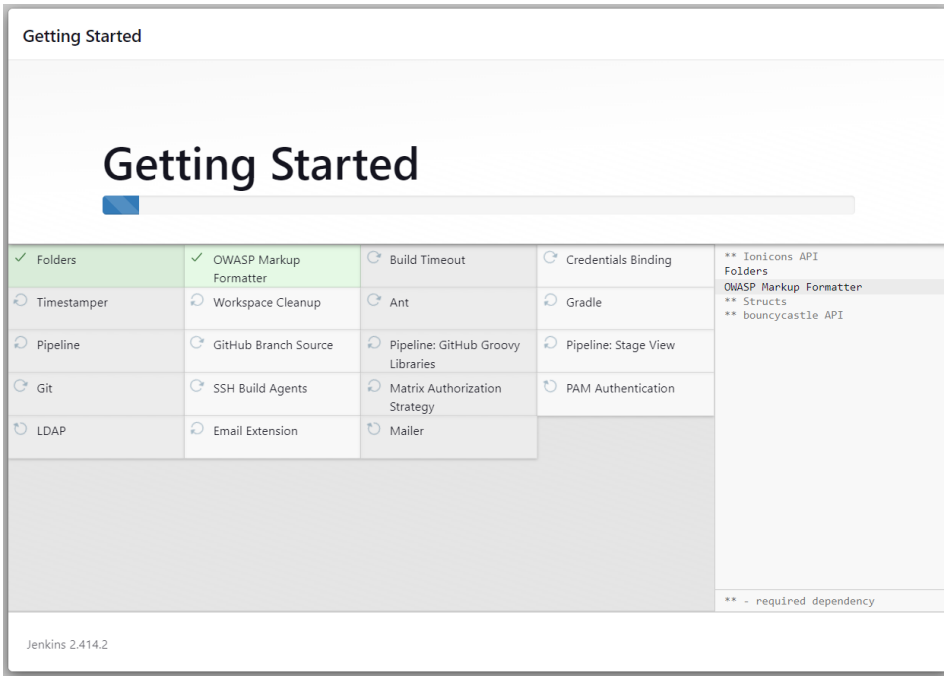
For example,

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Pw:

ea89c8fa77e44aa68d3d5fdc121f541f





Integrating Jenkins with GitHub

Install git

https://git-scm.com/download/win

- First create a new job in Jenkins, open the Jenkins **Dashboard** and click on "**create new jobs**".
- Now enter the item name and select the job type. For example, item name is **javaTpoint**" and job type is "**Freestyle project**".
- Click on **OK**.
- Once you click OK, the page will be redirected to its project configuration. Enter the project information:
- Now, under the "Source Code Management" you will see the Git option, if your **Git** plugin has been installed in Jenkins:
- Enter the Git repository URL on the "Repository URL" option to pull the code from GitHub.

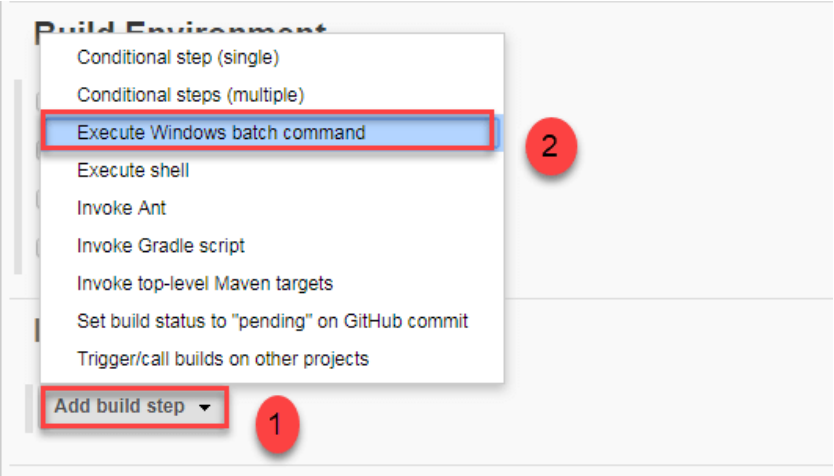
https://github.com/kriru/firstJava.git

Tweak the settings

Now that you have provided all the details, it's time to build the code. Tweak the settings under the **build** section to build the code at
the time you want. You can even schedule the build to happen periodically, at set times.

Under **build**,

1. Click on "**Add build step**"
2. Click on "**Execute Windows batch command**" and add the commands you want to execute during the build process.

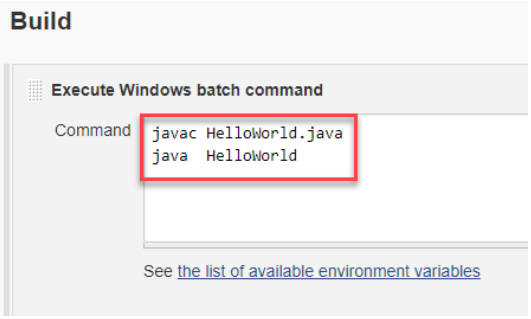


Here, I have added the java commands to compile the java code.

I have added the following windows commands:

javac HelloWorld.java

java HelloWorld



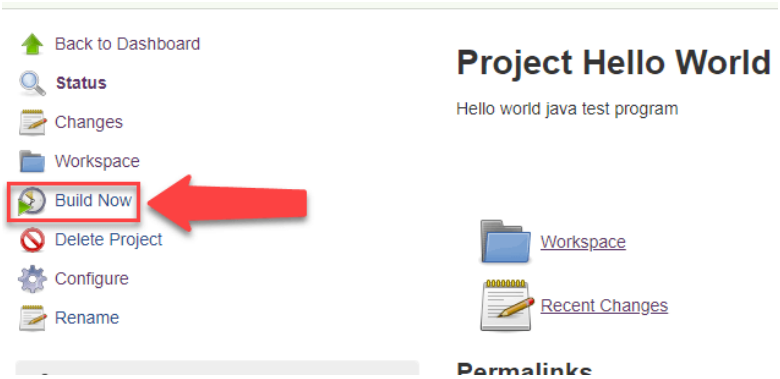
Save the project

When you have entered all the data,

1. Click **Apply**
2. **Save** the project.

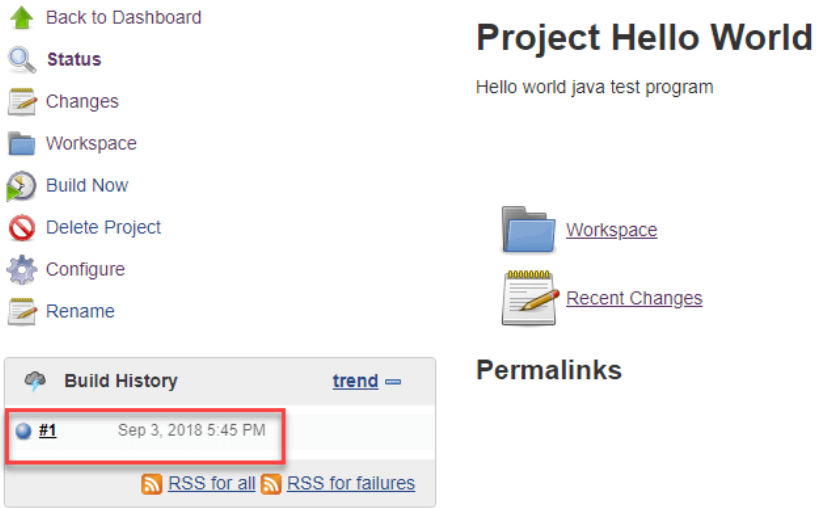
Build Source code

Now, in the main screen, Click the **Build Now** button on the left-hand side to build the source code.



Check the status

After clicking on **Build now,** you can see the status of the build you run under **Build History**.



See the console output

Click on the **build number** and then Click on **console output** to see the status of the build you run.

It should show you a success message,



In sum, we have executed a HelloWorld program hosted on GitHub. Jenkin pulls the code from the remote repository and builds continuously at a frequency you define.

Using Jenkins to build and inspect code

Jenkins - Code Analysis

The Jenkins Next Generation plugin helps in analyzing code, collects all warnings, and helps in visualizing the records through pie charts and graphs.

- Install Warnings Next Generation plugin from Manage Jenkins → Manage Plugins



task configuration page will open. Open the list "**Add post-build action**"

at the bottom of this page in the item "**Post-build Actions**". In the list, select **"Record compiler warnings and static analysis results":**

Jenkins to inspect code using FindBugs, PMD, and Checkstyle

- Add new item just as before
- **Build-> invoke top level maven targets**
- **Goals-> clean compile findbugs:findbugs pmd:pmd checkstyle:checkstyle**
- Click apply

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- Add **post build->record compile warnings and static analysis results**
- Build now

